MIL-R-20387C
12 November 1975
SUPERSEDING
MIL-R-20387B
24 September 1964, and
MIL-R-20380B
29 September 1964

MILITARY SPECIFICATION

RULES, STEEL, MACHINIST'S:

ENGLISH AND ENGLISH-METRIC GRADUATIONS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of steel rules.

1.2 Classification. Rules shall be of the following types and styles, as specified (see 6.2):

Type I - Beveled on one edge; graduated in inches.

Style A - Graduated in 32nds of an inch; 36-inch length.
Style B - Graduated in tenths of an inch to the right of the zero mark and hundredths of an inch to the left of the zero mark; 26-inch length.

Type II - Beveled on two edges; graduated in 64ths of an inch and halves of a millimeter; 36-inch length.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

FSC 5210

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SPECIFICATIONS

Federal

QQ-S-766	- Steel Plates, Sheets, and Strip - Corrosion Resisting.
PPP-P-40	- Packing and Packaging of Hand Tools.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The rules shall be made of corrosion-resisting steel. Beveled edges shall have a minimum thickness of 0.025 inch, plus or minus 0.010 inch, with a width of bevel of not less than 5/16 inch. The working edges shall be straight to within 0.003 inch per foot of length and parallel to within 0.003 inch per foot of length. The ends of the rules shall be at right angles to within 0.003 inch for the required width. All surfaces shall be true planes to within 0.001 inch from the edges. Each rule shall have a 3/8-inch-diameter hole within 1 inch of the zero end for hanging.

3.2 Material. Material shall be as specified herein.

3.2.1 Corrosion-resisting steel. Corrosion-resisting steel shall conform to QQ-S-766, Class 420.

3.3 Graduations. Graduations shall be on the beveled edges of the rules and shall be machine divided. Graduation lines shall be straight, uniform, clear, and distinct, and shall not exceed 0.005 inch

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in width. The graduations shall be permanently cut or etched in the face of the rule. Graduations and numbering shall be arranged so that the beveled (graduated) edge becomes the upper edge away from the user when in use. Graduation lines of decreasing lengths shall represent fractional aliquots of each scale in descending order. Lines representing equal fractional aliquots shall be of equal length and distinguishable by their relative length from graduation lines representing other fractional aliquots. Accuracy of graduations shall be such that the maximum error in distance between any 2 graduation marks shall not exceed 0.002 inch at 70° to 80° F.

3.4 Type I rule. The Type I rules shall be beveled on one edge and graduated in inches.

3.4.1 Style A. The Style A rules shall be 36 inches long, 1-1/4 inches wide, and 3/64 inch thick, with a minus tolerance of 0.016 inch on any dimension. The rule shall be graduated over the entire length, subdivided into 8ths, 16ths, and 32nds. The inch marks shall be numbered to read from left to right. The rule shall be marked "1/32 inch" within 1 inch to the first graduation mark.

3.4.2 Style B. The Style B rules shall be 26 inches long, 2-1/4 inches wide, and 1/8 inch thick, with a minus tolerance of 0.016 inch on any dimension. The rule shall be graduated in inches to the right of the zero mark for a length of 24 inches, and to the left of the zero mark for a length of 1 inch. The graduations to the right of the zero mark shall provide a minimum reading of 1/10 inch and the inch marks shall be numbered to read from left to right. The graduations to the left of the zero mark shall provide a m inimum reading of 1/100 inch and shall be numbered every 1/10 of an inch to read from right to left.

3.5 Type II rule. The Type II rule shall be 36 inches long, 1-1/4 inches wide, and 1/16 inch thick, with a minus tolerance to 0.016 inch on any dimension. The rule shall be graduated in 64ths of an inch and halves of a millimeter and shall be beveled on two edges. One beveled edge shall be graduated in centimeters subdivided into millimeters and halves of millimeters. The inch marks and the centimeter marks shall be numbered to read from left to right. The inch and millimeter scales shall be marked "1/64 inch" and "1/2 millimeter", respectively, within 1 inch of the first graduation mark.

3.6 Identification marking. The rules shall be identified in accordance with MIL-STD-130.

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3.7 Finish. The rules shall have a standard polish, nonglare finish, except that cadmium plating shall not be used on the rule (see 6.3).

3.8 Workmanship. The rule shall be free from defects such as sharp edges, burrs, cracks, dents, tool marks, and other defects that could impair the serviceability of the rule.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105, Inspection Level II.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined as specified in 4.4.1. AQL shall be 4.0 percent defective.

4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2.1 through 4.4.2.5. AQL shall be 4.0 percent defective. Failure of any test shall be cause for rejection.

4.4 Inspection procedure.

4.4.1 Examination. The rules shall be examined as specified herein for the following major defects:

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Custodians:

Army - ME
Air Force - 84

Review activity:

Army - MU

User activity:

Army - GL

Preparing activity:

Army - ME

Project No. 5210-0218

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